

GPU Sharing Mechanisms for Multi-Tenant Inference Serving

Systematic Literature Review Protocol

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27 September 2026

This protocol specifies the review methods in advance of final study selection. Protocol reporting follows PRISMA-P [1]; the completed review will use PRISMA 2020 [2]. Search documentation follows PRISMA-S [3] and software-engineering review guidance [4].

1. Research objective

Identify, critically appraise and synthesize peer-reviewed primary studies evaluating how **one physical GPU on a single node** is shared by multiple inference tenants. Compare temporal time-slicing, CUDA Multi-Process Service (MPS), Multi-Instance GPU (MIG), CUDA/GPU API remotng or virtualization, and hybrids. The intended output is a conditional engineering choice: which mechanism fits a specified model and arrival pattern, GPU generation/configuration, latency service-level objective (SLO), utilization or tenant-density goal, and isolation requirement. Treat performance interference, resource/memory separation, security isolation and fault containment as distinct observations; no mechanism name proves a measured outcome [10], [11].

2. Research questions

1. **RQ1 (alternatives):** Which of the four assigned mechanism families and hybrids have been evaluated for concurrent inference tenants on a single GPU, and what control/resource boundaries were evaluated?
2. **RQ2 (measurement):** How are utilization, throughput/goodput, tenant density, p50/p95/p99 latency, SLO violations, interference and isolation measured? Which studies can actually be compared?

3. **RQ3 (conditions):** How do workload/model, request arrival and batching, tenant mix, GPU generation, MPS settings, MIG profile and remoting path affect the reported outcomes?
4. **RQ4 (trade-offs):** What performance, isolation, implementation and operational trade-offs, conflicting findings and evidence gaps are reported?
5. **RQ5 (decision):** Under stated SLO, utilization/density, isolation and hardware conditions, what mechanism or combination is supported, and how strong is that support?

Each question will be linked to the extraction fields in §11 and synthesis outputs in §12.

3. Selected academic databases

The four selected academic sources are **IEEE Xplore** (engineering and systems), **ACM Digital Library** (computing and systems), **ScienceDirect** (Elsevier engineering and computing journals), and **Springer Nature Link** (computing journals and proceedings). NED's E-Resources page lists these platforms among its E-Databases [5]. Their institutional access, exact collection scope and export route will be verified before the searches; an unavailable source will be replaced only with a documented coverage rationale.

Scopus and Web of Science may supplement cross-publisher coverage and citation checking if accessible; any systematic runs will be logged and screened under the same criteria. Google Scholar and backward/forward references are supplementary discovery routes, logged separately. Coverage of systems proceedings such as USENIX will be checked because publisher collections do not index every venue.

4. Search keywords

Concept	Keywords and spelling variants
Hardware	GPU; graphics processing unit; accelerator
Inference	inference; model serving; deep learning; DNN; neural network; LLM; online serving
Temporal sharing	time slicing; time-slicing; time sharing; context switching; preemptive scheduling
MPS	MPS; multi-process service; CUDA MPS; process concurrency
MIG/partitioning	MIG; multi-instance GPU; GPU partitioning; spatial sharing
API remoting	API remoting; CUDA remoting; GPU proxy; API interception; virtual GPU; GPU virtualization
Outcomes/setting (optional refinements)	multi-tenant; co-location; utilization; latency; tail latency; QoS; SLO; interference; isolation

The mandatory query logic is hardware AND inference AND an OR of **all four** mechanism families. Do not require multi-tenant or an outcome word in every record: relevant seed papers may lack that literal term [12]. Spelling, tokenization, wildcard and subject-field choices will be checked in each

interface before formal searches. MPS and MIG alone can have unrelated meanings, so pair with GPU/inference and inspect precision.

5. Search strings

Master Boolean search string (translated into each database's accepted syntax):

(GPU OR "graphics processing unit")
 AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM)
 AND ("time slicing" OR "time sharing" OR "context switching"
 OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU"
 OR "GPU partitioning" OR "API remoting" OR "CUDA remoting"
 OR "GPU proxy" OR "GPU virtualization")

Academic database	Search implementation and fields
IEEE Xplore Command Search	Use Command Search with uppercase Boolean operators; record grouping, metadata fields, date/type facets and the executed query.
ACM DL Advanced Search	Use Advanced Search in the ACM full-text collection ; record generated syntax, field scope and collection (rather than assuming the broader ACM Guide).
ScienceDirect Advanced Search	Use Advanced Search; record accepted syntax, searched fields and article/date filters.
Springer Nature Link Advanced Search	Use Advanced Search; record phrase and Boolean behavior, field scope, year/content-type facets and the executed query.

If an interface limits clauses or misses known relevant terminology, split the strategy into **four separately logged searches per database**, using the common GPU AND inference blocks and one mechanism-family block from §4 each. Deduplicate exported records afterward. PipeSwitch [12], ParvaGPU [13] and Torpor [14] are examples for checking temporal, partitioning and remoting terminology, respectively; their citations here do not determine eligibility. A source absent from a publisher collection is a coverage issue rather than necessarily a query-syntax failure.

For every database run, record the collection, exact executed query, fields, filters, date/time, URL, total hits, export filename and reviewer. Record syntax tests separately from formal searches in accordance with PRISMA-S [3]. Backward and forward citation searches begin after at least one eligible full text is identified; record seed, direction, date, source and candidate IDs and apply the same screening criteria.

6. Publication period

Search for English-language works dated **1 January 2020 through 27 September 2026 inclusive**. Record the actual date on which each database is searched. If an interface filters by year only, inspect 2026 records and exclude post-cutoff publications manually. An older seminal mechanism

or indispensable baseline may enter only with a per-study written reason and a demonstrated direct connection to the RQs; older contextual work is otherwise background, not part of the included set. The review targets 15–20 **unique included primary studies**, with 15 as the required minimum. Eligibility will not be relaxed simply to reach a target count.

7. Inclusion criteria

All conditions must hold after full-text assessment:

1. English, peer-reviewed journal or full conference paper reporting an original empirical or technical evaluation; ordinarily within §6.
2. A clearly identifiable local GPU sharing, partition, process-concurrency or API-interception/remoting mechanism involving at least two inference tenants, services, models or processes on a physical GPU. A transferable mixed workload qualifies only if inference-specific evidence and transfer rationale are explicit.
3. Sufficient full-text detail to identify workload, sharing configuration and at least one RQ-relevant engineering outcome or directly assessable isolation property.
4. A unique study, with multiple conference/journal/preprint reports linked and the most complete peer-reviewed report selected for extraction.

Eligibility is independent of whether a mechanism performs well. Report negative and null findings. Record the rationale and reviewer decision for any older exception.

8. Exclusion criteria

At full text record one primary reason, with secondary notes if useful: **E1** no inference-serving evidence; **E2** no relevant single-node multi-tenant sharing; **E3** mechanism outside assigned scope; **E4** training-only; **E5** cluster orchestration without local sharing evaluation; **E6** no peer-reviewed original primary study (survey, vendor page, blog, preprint-only, poster, editorial, patent); **E7** insufficient technical/outcome evidence; **E8** duplicate/less complete report of the same study; **E9** outside date/language rule without exception; **E10** other, with written justification. Vendor guides [10], [11] describe mechanism context but cannot count as primary evidence or demonstrate workload performance. Track full text **not retrieved** as an access outcome, with attempts, rather than an exclusion among assessed reports [2].

9. Screening procedure

1. **Identification:** Execute and export each frozen query. Assign a unique record ID to every hit, preserving database/run provenance. Log citation-chased candidates separately.
2. **Deduplication and versions:** Normalize DOI, title, year and authors; retain raw records and link record IDs → report IDs → underlying unique study IDs. Prefer the complete peer-reviewed version without treating its preprint as a second study.
3. **Calibration:** Both screeners independently apply the draft rule set to a small common pilot set; record disagreements and clarify ambiguous terms before freeze. A third reviewer

adjudicates.

4. **Title/abstract:** Two independent screeners mark include, exclude or uncertain with reason and IDs. Retrieve uncertain records; adjudicate disagreement without inventing a consensus.
5. **Full text:** Record retrieval attempts and access status. Two screeners independently apply §§7–8 to retrieved reports; the third adjudicates discrepancies and records the final reason and study/report links. No inference from an unavailable full text.
6. **Flow:** Derive PRISMA 2020 record, report and study counts from those logs, including pre-screen removals, reports sought, reports not retrieved, assessed reports, exclusions with reasons and included unique studies [2]. Reconcile counts by formal and supplementary discovery route.

If the search cannot produce the required minimum of 15 eligible studies, seek instructor guidance before changing the scope or criteria.

10. Quality-assessment criteria

Two reviewers independently score each **included** study on Q1–Q10: **1** adequate, **0.5** partial/unclear, **0** absent/not reported. Reconcile with a recorded rationale; the third member resolves persistent disagreement. The rubric will be applied consistently to all included studies [4].

ID	Question
Q1	Is the engineering question and tenant setting explicit?
Q2	Are the mechanism and isolation/configuration described sufficiently?
Q3	Are GPU SKU/count, partition, OS, driver and runtime stated?
Q4	Are model, workload, batching/arrivals and co-tenant mix stated?
Q5	Is the baseline appropriate and configured fairly?
Q6	Are utilization, latency percentiles and other metric definitions/units clear?
Q7	Are warm-up, duration, repetitions/variance and load generation adequate?
Q8	Do figures/tables/data support the reported findings?
Q9	Are limitations, confounders and validity threats discussed?
Q10	Is there enough information to reproduce or critically interpret the comparison?

Total 0–10: **high 8–10, moderate 6–7.5, low <6**. Score does not replace eligibility. Retain eligible low-quality evidence with labels and rerun conclusions without it. Log reviewer-specific scores, evidence anchors, reconciliation and missing facts.

11. Data-extraction fields

Use one study-level identity row plus report/result rows where needed, each with PDF page/section/table/figure anchors. Test the form on examples from multiple mechanism families before extraction.

Field group	Required fields
Provenance	Record/report/study IDs, query/run and database, DOI/publisher URL, authors, title, year, venue, peer-review/version status, access/retrieval date, source anchor
Mechanism	Time-slice policy or context switch; MPS version/limits; MIG profile; API remotng/proxy path; hybrid; scheduler/control layer, admission/QoS policy
Environment	GPU vendor/SKU/count/memory/partition, host CPU/RAM, OS/kernel, driver, CUDA/runtime, container or VM
Workload	Inference task/model/size/input, tenant identity and mix, batch size, arrival pattern, concurrency, load, warm-up/window/repetitions
Comparator	Baseline, direct vs indirect comparison, configuration equivalence
Outcomes	Utilization/occupancy, throughput/goodput, tenant density, p50/p95/p99 and units, SLO threshold/miss rate, interference, overhead, energy if given, uncertainty/variance
Isolation	Separately measured or claimed performance interference, resource/memory partitioning, security boundary and fault containment; mark not reported where absent
Interpretation	Verbatim-sized finding summary in own words, exact anchor, author caveat, reviewer inference separately, QA scores, confounders and comparability notes

Do not fill absent values by assuming a GPU mechanism's default behavior. Capture both favorable and unfavorable findings.

12. Evidence-synthesis strategy

Build a mechanism taxonomy (temporal, MPS, MIG, API remotng/virtualization, hybrids), an RQ-to-study matrix and a condition/comparability table. For RQ1 summarize evaluated control and isolation dimensions; for RQ2 compare metric and experiment definitions; for RQ3 stratify by GPU generation/profile, model and batch/arrival/tenant conditions; for RQ4 trace reported trade-offs and counterexamples to differing baselines and conditions; for RQ5 create a conditional decision matrix with evidence strength and uncertainty.

Use structured narrative synthesis because different GPUs, workloads, tenancy and latency percentiles make raw pooling misleading. Compare numerical effects only inside genuinely compatible experimental strata; otherwise report direction, conditions and limitations without an invented aggregate. Report each recommendation as **direct within-study comparison**, **indirect primary**

evidence, vendor context only, or insufficient evidence. Apply the SWiM transparency principles for any synthesis without meta-analysis [9]. Repeat the conclusion after excluding low-QA/incomplete studies. Report coverage gaps, publication/access bias, heterogeneity and reviewer disagreement. No mechanism is declared universally optimal.

Protocol amendments

Any major change after searching begins—such as a change to databases, search logic, date range, eligibility, screening, quality assessment, extraction or synthesis—will be entered in a dated amendment log. Each entry will describe the previous and revised rule, the reason for change, approving reviewers, affected searches and records, any required reruns, and the effect on screening and PRISMA counts. The report will disclose these deviations and their justification [1].

References

- [1] L. Shamseer *et al.*, “Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation,” *BMJ*, vol. 349, g7647, 2015. doi: [10.1136/bmj.g7647](https://doi.org/10.1136/bmj.g7647).
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- [3] M. L. Rethlefsen *et al.*, “PRISMA-S: an extension to the PRISMA statement for reporting literature searches in systematic reviews,” *Systematic Reviews*, vol. 10, art. 39, 2021. doi: [10.1186/s13643-020-01542-z](https://doi.org/10.1186/s13643-020-01542-z).
- [4] B. Kitchenham and S. Charters, “Guidelines for performing systematic literature reviews in software engineering,” EBSE Technical Report EBSE-2007-01, 2007. [Online]. Available: [EBSE record](#).
- [5] Engr. Abul Kalam Library, NED University, “E-Resources,” accessed Sept. 27, 2026. [Online]. Available: [NED E-Resources](#).
- [6] IEEE, “IEEE Xplore: Searching and Saving Searches,” user guide. [Online]. Available: [IEEE guide](#).
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- [9] M. Campbell *et al.*, “Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline,” *BMJ*, vol. 368, 16890, 2020. doi: [10.1136/bmj.16890](https://doi.org/10.1136/bmj.16890).
- [10] NVIDIA, “Introduction,” *Multi-Instance GPU User Guide*. [Online]. Available: [MIG introduction](#).
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- [12] Z. Bai *et al.*, “PipeSwitch: Fast pipelined context switching for deep learning applications,” in *Proc. 14th USENIX OSDI*, 2020. [Online]. Available: [USENIX venue page](#).
- [13] M. Lee *et al.*, “ParvaGPU: Efficient spatial GPU sharing for large-scale DNN inference in cloud environments,” in *Proc. SC '24*, 2024. doi: [10.1109/SC41406.2024.00048](#).
- [14] M. Yu *et al.*, “Torpor: GPU-enabled serverless computing for low-latency, resource-efficient inference,” in *Proc. USENIX ATC '25*, 2025. [Online]. Available: [USENIX venue page](#).